

Technical Validation Report: Private Local-First Document Intelligence System

Prepared for: Solo Attorney

Subject: Validation and Refinement of CE_PLAN.md

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Executive Summary

Your proposed plan for a private, local-first document intelligence system (CE_PLAN.md) is structurally sound, legally defensive, and technologically viable in 2026. Your core instincts—mandating local-only execution for attorney-client privilege, establishing a strict boundary between retrieval and agentic action, and insisting on page-level citation verification—are precisely aligned with the latest empirical research on legal AI.

However, a review of recent academic benchmarks, community learnings, and hardware markets reveals several critical vulnerabilities in the original plan. Most notably, the assumption that Retrieval-Augmented Generation (RAG) effectively neutralizes hallucinations is contradicted by 2025 Stanford research showing that even commercial legal RAG tools hallucinate up to 33% of the time ¹. Furthermore, legal documents present unique retrieval challenges, such as Document-Level Retrieval Mismatch (DRM), where a system correctly identifies a relevant clause but pulls it from the wrong client's contract ².

This report validates your core architecture while providing critical course corrections regarding hardware selection, embedding model evaluation, and citation mechanics. It concludes with a revised implementation roadmap and a concrete acceptance checklist.

1. Critical Plan Red Flags and Mitigations

The original CE_PLAN contains several assumptions that must be adjusted to ensure the system is reliable for legal practice.

The "Hallucination-Free" RAG Myth

The most significant risk in the current plan is the implicit trust placed in RAG to ensure factual accuracy. A landmark 2025 study by Stanford University evaluated leading commercial legal RAG tools (Lexis+ AI, Westlaw AI-Assisted Research, and Ask Practical Law AI) and found that they hallucinated between 17% and 33% of the time ¹. The researchers defined a hallucination not just as generating false information, but also as falsely asserting that a source supports a proposition ¹.

Mitigation: RAG reduces, but does not eliminate, hallucination. Your requirement that the attorney remains responsible and must verify every citation is non-negotiable. To enforce this mechanically, the system should implement span-level citation verification, requiring the LLM to cite specific line ranges or offsets that must mechanically overlap with the retrieved evidence 3 .

Document-Level Retrieval Mismatch (DRM)

Legal documents are highly redundant. Standard vector search struggles to differentiate between identical boilerplate clauses found in different contracts. This leads to Document-Level Retrieval Mismatch (DRM), a catastrophic failure mode where the system retrieves the correct legal language but from an entirely unrelated source document 2 .

Mitigation: The retrieval pipeline must utilize strong metadata filtering (e.g., scoping searches strictly by matter ID, client, or date) before vector similarity is calculated. Furthermore, implementing Summary-Augmented Chunking (SAC)—where a document-level synthetic summary is prepended to each chunk—injects necessary global context and has been shown to significantly reduce DRM 2 .

The MTEB Leaderboard Trap

The plan suggests relying on top-performing models from the Massive Text Embedding Benchmark (MLEB). However, general MTEB rankings do not correlate with legal retrieval accuracy. The Massive Legal Embedding Benchmark (MLEB) demonstrates that domain adaptation is far more critical; general-purpose models that dominate MTEB often underperform on legal-specific tasks compared to specialized models 4 .

Mitigation: You must evaluate local embedding models (such as qwen3-embedding or bge-m3) against your own "golden set" of legal queries and documents. Do not select an embedding model based solely on public leaderboards. Additionally, employing hybrid retrieval (combining dense vector search with sparse/BM25 keyword search) is highly recommended to accurately capture specific legal terms of art 5 .

2. Hardware Strategy (2026 Landscape)

The original plan references the RTX 4080/4090 class of GPUs. In 2026, the NVIDIA 40-series has been discontinued, and the 50-series (Blackwell) has been introduced amid a severe GDDR7 memory shortage that has inflated prices 6 . For a single attorney performing retrieval and summarization on a corpus of a few thousand documents, a 70-billion parameter model is unnecessary and adds prohibitive cost and latency. The sweet spot is a 14B to 32B parameter model running on 16GB to 32GB of VRAM 7 .

Option A: The "Low-Maintenance Appliance" (Recommended)

For a law office prioritizing silence, low power consumption, and zero Windows administration, Apple Silicon is the optimal choice. A **Mac Studio M4 Max with 64GB to 128GB of unified memory** (approximately \$3,500 to \$5,000) provides 546 GB/s of memory bandwidth ⁸. This system can comfortably run 14B models at high speeds and can even accommodate 70B models at 8 to 15 tokens per second if necessary ⁸. It serves as a reliable, single-box appliance.

Option B: The CUDA Performance Path

If maximum tokens-per-second and lowest latency are required, or if you plan to scale to a multi-user environment using vLLM, an NVIDIA workstation is appropriate. A PC equipped with an **RTX 5070 Ti or 5080 (16GB VRAM)** or an **RTX 5090 (32GB VRAM)**, paired with 64GB of system RAM and fast NVMe storage, will cost between \$5,000 and \$8,000 ⁶. The 16GB cards handle up to 24B-32B models efficiently using 4-bit quantization, while the 32GB 5090 provides substantial headroom ⁷.

3. Recommended Software Stacks

The choice of software depends on whether you are validating the concept (Pilot) or building a hardened, citation-grade system (Production).

The Pilot Stack (Turnkey Validation)

To prove the value of the system quickly without writing custom code, deploy this stack on your existing hardware.

Component	Recommendation	Rationale
Frontend / RAG	AnythingLLM	In a May 2026 benchmark of 5,000 pages, AnythingLLM demonstrated the lowest hallucination rate (6%) and provided the best out-of-the-box citations (filename plus page number, clickable inline) ⁹ . It avoids the citation shortcomings of Open WebUI.
Runtime	Ollama	The simplest local serving engine, providing an OpenAI-compatible API and easy model management ¹⁰ .

LLM	Qwen3 14B (or 8B)	Widely recognized in 2026 as the best all-around local model for RAG, offering strict instruction following and structured output 11.
Embedding	bge-m3	Supports hybrid dense and sparse retrieval out of the box, which is crucial for legal terminology 12.

The Production Stack (Citation-Grade Custom Build)

For the final deployment, a custom orchestration layer is required to enforce mechanical citation verification and mitigate DRM.

Component	Recommendation	Rationale
Orchestration	Custom Python API (LlamaIndex)	LlamaIndex excels at complex data indexing, hierarchical chunking, and response synthesis, making it the superior framework for a controllable, citation-grade build 13.
Vector DB	Qdrant or LanceDB	Qdrant offers best-in-class metadata filtering (essential for matter scoping) 14. LanceDB is an embedded alternative that requires no separate server, simplifying deployment 15.
LLM	Qwen3 14B or Mistral Small 3.1 24B	Mistral-family instruct models are highly disciplined in strict grounded RAG scenarios, degrading gracefully when evidence is missing 10.

Chunking & OCR	Docling + PyMuPDF	Docling (IBM) provides superior structure-preserving parsing (97.9% complex-table accuracy), outputting unified markdown that retains document hierarchy ¹⁶ .
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4. Revised Implementation Roadmap

This revised roadmap emphasizes rapid validation before significant capital expenditure.

Phase 1: Pilot & Validation (Weeks 1-3)

Goal: Prove retrieval quality and citation accuracy on a real matter using existing hardware.

- Install Ollama and AnythingLLM on your current workstation.
- Load a small, sanitized corpus from a closed matter.
- Build a "golden evaluation set" of 20 to 50 real legal questions mapped to known source pages.
- Measure the hallucination rate and citation fidelity. Do not proceed to Phase 2 until the system proves it can accurately retrieve and cite information.

Phase 2: Hardware Procurement & Infrastructure (Weeks 4-6)

Goal: Establish the physical and foundational software infrastructure.

- Procure the Mac Studio M4 Max or RTX 50-series workstation based on Phase 1 learnings.
- Implement full-disk encryption and establish a secure, offline backup protocol (e.g., external encrypted drives).
- Deploy the local OCR pipeline (PyMuPDF for routing, Docling for structure, Tesseract for fallback scans) and the chosen Vector DB (Qdrant or LanceDB).

Phase 3: Custom Orchestration & DRM Mitigation (Weeks 7-10)

Goal: Build the custom, citation-grade RAG layer.

- Develop the LlamaIndex pipeline.
- Implement hybrid retrieval (dense + BM25) and a cross-encoder reranker (e.g., `bge-reranker-v2-m3`).

- Implement Summary-Augmented Chunking (SAC) and strict metadata filtering to eliminate Document-Level Retrieval Mismatch (DRM).
- Enforce mechanical citation verification, requiring the LLM to output exact text offsets that overlap with the retrieved chunks.

Phase 4: Production Rollout & Workflow Integration (Weeks 11-12)

Goal: Safely integrate the system into daily legal practice.

- Finalize the user interface (a simple chat interface connected to the custom API).
- Train on the verification workflow: the attorney must click every citation and verify the source text before using the output.
- Strictly maintain the agentless boundary; the system must never draft and file documents autonomously.

5. Acceptance Checklist

Before relying on the system for active client matters, it must pass the following criteria:

- ☐ **Golden Eval Set Validation:** The system correctly answers 50+ real legal questions, with every answer mapped to the exact known source page.
- ☐ **Refusal Discipline:** The system must correctly refuse to answer when the required evidence is not present in the corpus (0% hallucination rate on "not found" questions).
- ☐ **Citation Fidelity:** 100% of generated factual claims must cite a specific filename and page/section.
- ☐ **DRM Resilience:** The system correctly distinguishes between identical boilerplate clauses located in different contracts using metadata filters.
- ☐ **Performance Metrics:** Inference latency is under 3 seconds to the first token, and the system does not encounter Out-Of-Memory (OOM) errors when processing a 32K context window.
- ☐ **Air-Gapped Security:** Network monitoring (e.g., Wireshark) confirms absolutely zero outbound data transmission during inference and retrieval operations.

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